

Trial Test 1

- Submit your answers via Google Forms (you have only 1 hour and 10 mins to complete).
- Your score will be available immediately after the test. Please, check your emails.
- If you have technical issues, let me know in DM or in the Group Chat.
- You are not allowed to use the calculator. Also, do not google, be honest!
- Do not panic and manage your time wisely. Read the questions carefully and try to answer all of them.

Good Luck!

1) Simplify:

$$5 * [5 * (6^2 - 5 * 3) + \sqrt{400}]^{1/3} + 7$$

- a) 0
- b) 25
- c) 56
- d) 32

2) Find the turning point of the parabola:

$$f(x) = 6x^2 + 12x + 3$$

- a) (-1,-3)
- b) (-2,-3)
- c) (1,3)
- d) (-2,3)

3) Solve the inequality:

$$\frac{1}{2} (3 + 4t) \leq 6\left(\frac{1}{3} - \frac{1}{2}t\right) - \frac{1}{4} (2 + 10t)$$

- a) [7.5, +∞)
- b) (-∞, 0]
- c) [0, +∞)
- d) [-7.5, +∞)

4) The roots of the equation $3x^2 - 14x + c = 0$, differ by 4.

Find the value of c :

a) $\frac{13}{9}$

b) $\frac{13}{3}$

c) $\frac{10}{9}$

d) $\frac{14}{3}$

5) How much pure alcohol has to be added to 400 ml of a solution containing 15% alcohol to change the concentration of the alcohol in the mixture to 32%?

a) 100 ml

b) 150 ml

c) 120 ml

d) 115 ml

6) The tenth term of an arithmetic sequence is $\frac{55}{2}$ and the second term is $\frac{7}{2}$.

Find the first term.

a) $\frac{3}{2}$

b) $\frac{1}{5}$

c) $\frac{12}{5}$

d) $\frac{1}{2}$

7) Ben is producing a scale drawing of a swimming pool he plans to build. The pool's dimensions are 2 m by 6 m. When he measures the drawing on his plan he sees that it is 8 cm by 24 cm.

What scale should he include in his plan?

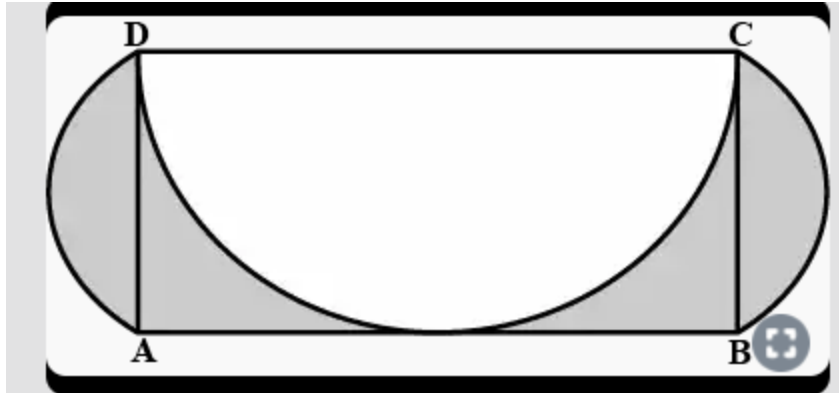
a) 1:10

b) 1:15

c) 1:25

d) 1:5

8) In the figure given below, ABCD is a rectangle. AB = 14 cm and BC = 7 cm. Taking DC, BC, and AD as diameter, three semicircles are drawn as shown in the figure. Find the area of the shaded portion.



a) 26.5

b) 59.5

c) 67.5

d) 43.5

9) Find the area of a parallelogram with vertices at points $(-6,-5)$, $(-2,4)$, $(5,4)$, and $(1,-5)$.

a) 63

b) 68

c) 56

d) 72

10) It is given that x is inversely proportional to the square root of y . When $x=2$, $y=16$.

What is the estimated value of y when $x=1024$?

a) 2^{-14}

b) 2^{-12}

c) 2^7

d) 2^{16}

11) Rearrange to make x subject

$$\frac{7x+10}{9x+5} = 3y^2 + 2$$

a) $\frac{15y^2}{7-9(3y^2+2)}$

b) $\frac{15y^2}{7+9(3y^2+2)}$

c) $\frac{-15y^2}{7-9(3y^2+2)}$

d) $\frac{-15y^2}{7+9(3y^2+2)}$

12) At a cinema, drinks are sold in regular and large sizes. The cups are mathematically similar. The ratios of depths and heights are 4:5. The volume of drink in a regular size cup is 320 cm^3 .

What is the volume of the large cup?

a) 640

b) 576

c) 625

d) 500

13) Q is 5 km away from P on a bearing of 065. R is 5 km away from Q on a bearing of 155.

What is the bearing of P from R?

a) 070

b) 290

c) 225

d) 270

14) PQR is an isosceles triangle in which $PQ=PR=6$ cm and $QR=8$ cm.

What is the value of the tangent of angle PQR?

a) $\frac{2}{3}$

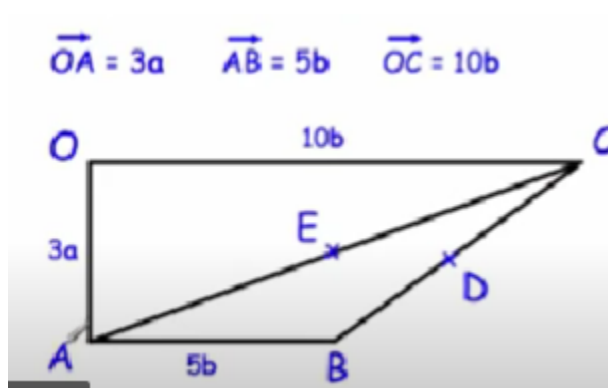
b) $\frac{3}{2}$

c) $\frac{2}{\sqrt{5}}$

d) $\frac{\sqrt{5}}{2}$

15) OABC is a trapezium. Point D is the midpoint of the BC and point E is the midpoint of AC.

Write the vector AE in terms of a and b.



a) $\frac{5}{2}a-5b$

b) $-\frac{3}{2}a-5b$

c) $-\frac{3}{2}a+5b$

d) $\frac{5}{2}a+5b$

16) A city football club collected money for charity at all of its matches for a year. At the end of the year the total collected was divided among three charities, A, B and C, in the ratio $1:\frac{2}{3}:\frac{4}{5}$. Charity C received £3 000.

What was the total amount collected for charity during the year?

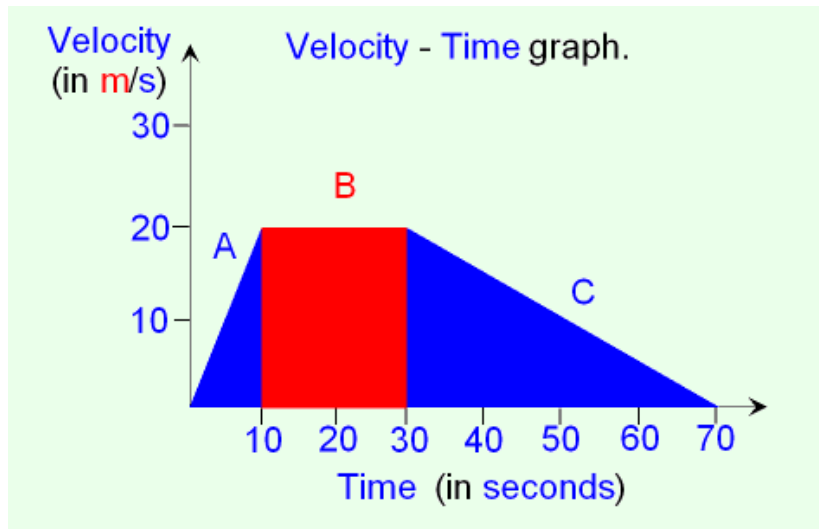
a) £9250

b) £8250

c) £9000

d) £7000

17) Calculate the total distance traveled by a traveler using the graph given:



- a) 900 m
- b) 600 m
- c) 1200 m
- d) 1000 m

18) PQRS is a square. Sides PQ and RS each are increased by 40%. Sides QR and PS are each decreased by 75%.

What is the total area decrease of the rectangle that has been obtained after these transformations?

- a) 10%
- b) 35%
- c) 65%

d) 70%

19) The mass of an atom of uranium is 4×10^{-25} kg.

What is the mass, in milligrams, of 8 million atoms of uranium?

a) 3.2×10^{-18}

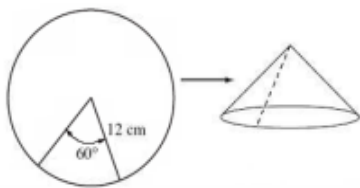
b) 3.2×10^{-19}

c) 3.2×10^{-12}

d) 3.2×10^{-14}

20) A 60 degree wedge is cut from a paper circle with radius 12 cm and the paper is then folded into a cone, as shown in the figure above.

What is the height, in cm, of this cone?



a) $\sqrt{44}$

b) $\sqrt{70}$

c) $\sqrt{108}$

d) $\sqrt{10\pi}$

21) The cross-section of a triangular prism is an equilateral triangle with side $2x$ cm. The length of the prism is d cm. Let the total surface area of the prism be $T \text{ cm}^2$.

Given that the volume of the prism is $T \text{ cm}^3$, which of the following is an expression of the d in terms of x ?

a) $\frac{x}{x+3}$

b) $\frac{3x}{3x-2\sqrt{3}}$

c) $\frac{2x}{x-2\sqrt{3}}$

d) $\frac{x}{x-2\sqrt{3}}$

22) The graph of $f(x) = x^3$ is translated 6 units up, 2 units right, and reflected over the x -axis. If the resulting graph represents $g(x)$, then $g(-1)$ is equal to

a) 21

b) 5

c) 14

d) 33

23) The equation $2x(ax - 4) - x^2 + 6 = 0$ has no real roots.

What is the smallest possible integer value of a ?

a) -1

b) 2

c) 4

d) 3

24) In the formula $P = 2r^2t$, $r = 3 * 10^{-3}$, $t = 2.5 * 10^4$.

Calculate the value of P . Leave your answer in standard form.

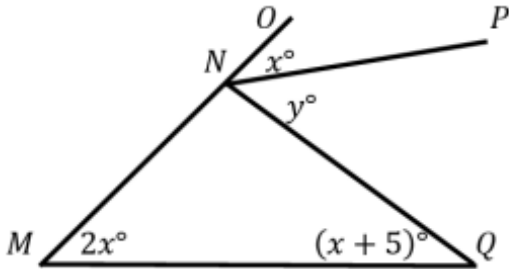
a) $7.5 * 10^{-2}$

b) $45 * 10^{-2}$

c) $75 * 10^{-2}$

d) $4.5 * 10^{-1}$

25) In the figure below, N lies on MO. In terms of x , y is given by



- a) $2x$
- b) $3x$
- c) $2x+5$
- d) $3x+5$

26) Convert 60 km/h to m/s

- a) 18.5
- b) 13.5
- c) 16.67
- d) 14.33

27) In a group of five friends, the sum of ages of each group of four of them are 124, 128, 130, 136 and 142.

What is the age of the youngest of the friends?

a) 21

b) 18

c) 16

d) 23

28) $\left(\frac{-2^2(-3)^3(-2)^2}{(-3)^2 3^{-2} 4^3}\right)=?$

a) $\frac{9}{2}$

b) $\frac{9}{4}$

c) $\frac{27}{2}$

d) $\frac{27}{4}$

29) The value of $\sqrt{56 + 24\sqrt{5}}$ is equivalent to

a) $12 + 2\sqrt{5}$

b) $3 + 4\sqrt{5}$

c) $6 + 2\sqrt{5}$

d) $1 + 12\sqrt{5}$

30) Solve

$$0,6(7)+0,(34)$$

a) $\frac{217}{220}$

b) $\frac{417}{330}$

c) $\frac{187}{220}$

d) $\frac{337}{330}$

